

Weight Initialization Methods in Deep Learning

Xavier / Glorot, He / Kaiming, and Orthogonal Initialization

Why initialization matters

Good initialization keeps activations and gradients in a useful range during training. Poor initialization can cause vanishing gradients, exploding gradients, or very slow learning.

1. Xavier / Glorot Initialization

Best suited for: sigmoid and tanh activations (and commonly used when the activation is approximately symmetric around zero).

The goal is to keep the variance of activations and back-propagated gradients approximately stable across layers. Let n_{in} be the number of inputs (fan-in) and n_{out} the number of outputs (fan-out).

Xavier/Glorot Normal: $W_{ij} \sim N(0, 2/(n_{in} + n_{out}))$

Xavier/Glorot Uniform: $W_{ij} \sim U(-a, a), \quad a = \sqrt{6/(n_{in} + n_{out})}$

Intuition: Xavier balances fan-in and fan-out. This is particularly useful for activations such as tanh, where preserving signal variance helps prevent saturation.

2. He / Kaiming Initialization

Preferred for: ReLU and ReLU-family activations such as Leaky ReLU. ReLU sets a substantial fraction of activations to zero, so He initialization compensates for this effect.

He Normal: $W_{ij} \sim N(0, 2/n_{in})$

He Uniform: $W_{ij} \sim U(-a, a), \quad a = \sqrt{6/n_{in}}$

Intuition: the factor 2 is designed to preserve activation variance through ReLU layers. For modern CNNs and many deep networks using ReLU-like activations, He initialization is a strong default.

3. Orthogonal Initialization

Orthogonal initialization starts from a random matrix and constructs an orthogonal (or semi-orthogonal) weight matrix. For a square matrix, the key property is:

$$W^T W = I$$

This means the columns of W are orthonormal. More generally, QR decomposition of a random matrix A can be used: $A = QR$, then Q is used as the initialized weight matrix (with appropriate scaling when needed).

Intuition: orthogonal transformations tend to preserve vector norms, which can help maintain stable signal propagation. They are especially useful in deep networks and recurrent settings where repeated transformations can otherwise amplify or shrink signals.

Quick Comparison

Method	Typical activation	Main variance rule	Key idea
Xavier / Glorot	sigmoid, tanh	$2 / (\text{fan-in} + \text{fan-out})$	Balance input and output variance
He / Kaiming	ReLU, Leaky ReLU	$2 / \text{fan-in}$	Compensate for ReLU's zeroed activations
Orthogonal	general / deep / recurrent	$W^T W = I$	Preserve vector norms / improve signal flow

Practical rule of thumb

- Using tanh/sigmoid → start with **Xavier / Glorot**.
- Using ReLU/Leaky ReLU → start with **He / Kaiming**.
- Need norm-preserving structure or recurrent/deep signal stability → consider **Orthogonal** initialization.

The exact best choice can depend on the architecture, activation, normalization layers, and optimizer.